

DS Assignment 03 by U24AI089

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
In [2]: df = pd.read_csv("../ML_5th_SEM_ASSIGNMENTS/titanic/train.csv")
df.head()
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0

```
In [5]: # Select numerical features
features = [
    "Age",
    "Fare",
    "SibSp",
    "Parch",
    "Pclass"
]

X = df[features]

X.head()
```

Out[5]:

	Age	Fare	SibSp	Parch	Pclass
0	22.0	7.2500	1	0	3
1	38.0	71.2833	1	0	1
2	26.0	7.9250	0	0	3
3	35.0	53.1000	1	0	1
4	35.0	8.0500	0	0	3

```
In [6]: ## Missing Values
X.isnull().sum()
```

Out[6]:

Age	177
Fare	0
SibSp	0
Parch	0
Pclass	0
dtype:	int64

```
In [7]: ## Handle missing values
X.fillna(X.median(), inplace=True)
X.isnull().sum()
```

```
C:\Users\rajde\AppData\Local\Temp\ipykernel_3792\4190253255.py:
2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a
DataFrame
```

```
See the caveats in the documentation:
https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-
copy
X.fillna(X.median(), inplace=True)
```

```
Out[7]: Age      0
Fare      0
SibSp     0
Parch     0
Pclass    0
dtype: int64
```

```
In [8]: ## Standardization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_scaled[:5]
```

```
Out[8]: array([[ -0.56573646, -0.50244517,  0.43279337, -0.47367361,
  0.82737724],
 [ 0.66386103,  0.78684529,  0.43279337, -0.47367361,
 -1.56610693],
 [-0.25833709, -0.48885426, -0.4745452 , -0.47367361,
  0.82737724],
 [ 0.4333115 ,  0.42073024,  0.43279337, -0.47367361,
 -1.56610693],
 [ 0.4333115 , -0.48633742, -0.4745452 , -0.47367361,
  0.82737724]])
```

```
In [12]: ## Apply PCA
pca = PCA()

X_pca = pca.fit_transform(X_scaled)

print("Before PCA:", X_scaled.shape)
print("After PCA:", X_pca.shape)
```

```
Before PCA: (891, 5)
After PCA: (891, 5)
```

Interpretation

Before PCA the features are like -

Age, Fare, SibSp, Parch, Pclass

After PCA without any dimensionality Reduction, they become -

PC1, PC2, PC3, PC4, PC5

```
In [16]: ## Explained Variance ratio
explained_variance = pca.explained_variance_ratio_

for i, variance in enumerate(explained_variance):
    print(
        f"PC{i+1}: {variance:.4f} "
        f"({variance * 100:.2f}%)"
    )
```

PC1: 0.3396 (33.96%)

PC2: 0.3252 (32.52%)

PC3: 0.1446 (14.46%)

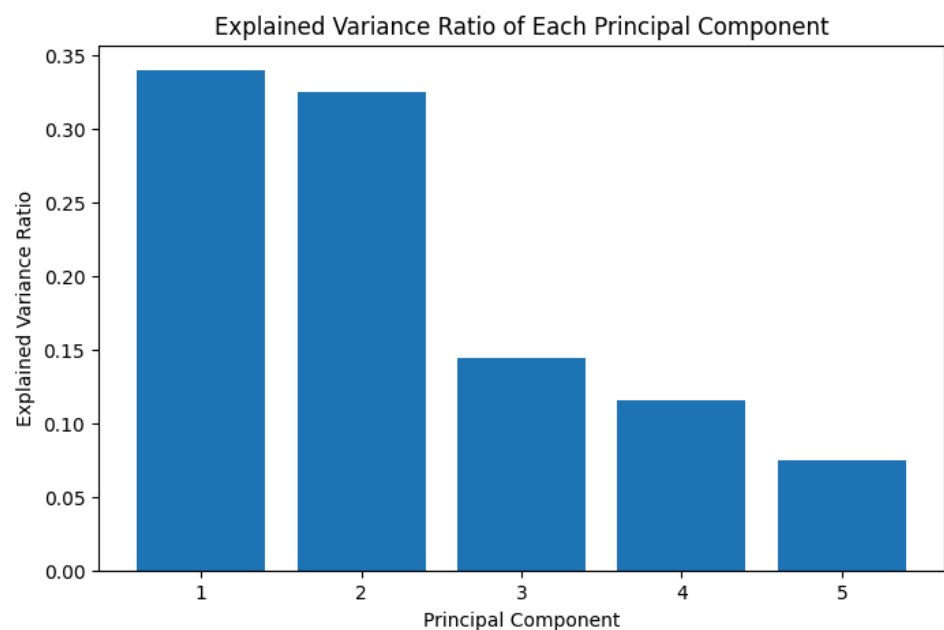
PC4: 0.1159 (11.59%)

PC5: 0.0748 (7.48%)

```
In [23]: ## Plot Explained Variance ratio
components = range(1, len(explained_variance) + 1)

plt.figure(figsize=(8, 5))
plt.bar(components, explained_variance)
plt.xlabel("Principal Component")
plt.ylabel("Explained Variance Ratio")
plt.title("Explained Variance Ratio of Each Principal Component")

plt.show()
```



```
In [26]: ## Reduce dimensions to 2
pca_2 = PCA(n_components=2)

X_pca_2 = pca_2.fit_transform(X_scaled)
print("Original shape:", X.shape)
print("After PCA:", X_pca_2.shape)
```

Original shape: (891, 5)

After PCA: (891, 2)

```
In [28]: ## Convert the 2 principal components into a DataFrame
pca_df = pd.DataFrame(
    X_pca_2,
    columns=["PC1", "PC2"]
)

pca_df.head()
```

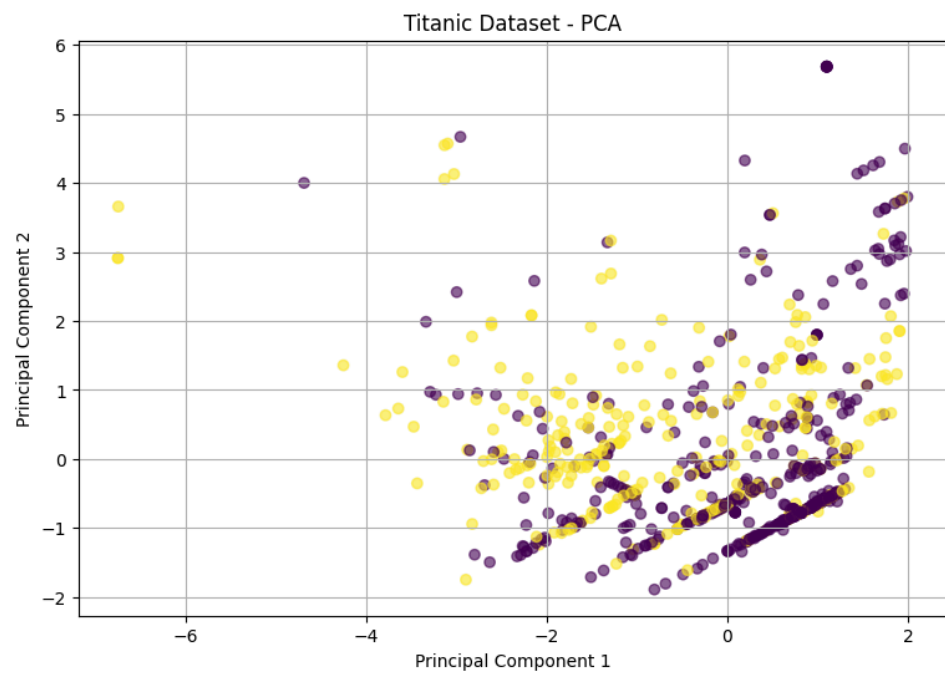
Out[28]:

	PC1	PC2
0	1.140660	-0.076520
1	-1.768445	0.133479
2	0.879196	-0.726606
3	-1.458103	0.071555
4	0.561252	-0.943713

```
In [34]: ## Visualize the 2 principal components
plt.figure(figsize=(9, 6))

plt.scatter(
    pca_df["PC1"],
    pca_df["PC2"],
    c=df["Survived"],
    alpha=0.6,
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Titanic Dataset - PCA")
plt.grid()
plt.show()
```



```
In [36]: ## Comparison Between Before and After PCA
print("===== BEFORE PCA =====")

print("Number of rows:", X.shape[0])
print("Number of features:", X.shape[1])
print("Features:", list(X.columns))

print("\n===== AFTER PCA =====")

print("Number of rows:", X_pca_2.shape[0])
print("Number of features:", X_pca_2.shape[1])
print("Features:", ["PC1", "PC2"])

===== BEFORE PCA =====
Number of rows: 891
Number of features: 5
Features: ['Age', 'Fare', 'SibSp', 'Parch', 'Pclass']

===== AFTER PCA =====
Number of rows: 891
Number of features: 2
Features: ['PC1', 'PC2']
```

```
In [ ]: ## Variance retained by PC1 and PC2
total_variance = pca_2.explained_variance_ratio_.sum()

print(
    f"Variance retained by PC1 and PC2: "
    f"{total_variance * 100:.2f}%"
)
```

Variance retained by PC1 and PC2: 66.48%